A CLOSED-FORM JOINT GAUSSIAN MIXTURE CONSTRUCTION OF A DIFFEOMORPHISM

ALI DARIJANI[⊛□#¶⊥] AND JÜRGEN BEYERER[⊛⊥]

Abstract. A closed-form construction of a joint Gaussian mixture induced by a diffeomorphism is derived. Given a diffeomorphism $f\colon U \subset \mathbb{R}^d \to X \subset \mathbb{R}^d$, a uniform probability density function over a partition of unity discretization is defined and $\mathcal{G}(u|g_U)$ is constructed. $\mathcal{G}(x|g_X)$ is constructed using $\mathcal{G}(u|g_U)$ and the Jacobian of the diffeomorphism. Finally, a joint $\mathcal{G}(u, x|g_U, g_X)$ is constructed, removing the need to work with the diffeomorphism explicitly. All marginals and conditionals of the resulting joint distribution can then be evaluated in closed form, removing the need to explicitly access the diffeomorphism. As an example, a particle moving along a segment of a circular trajectory is modeled probabilistically, demonstrating that position can be inferred from velocity directly, without accessing the underlying time parameter.

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Definition 0.1 (Closed-form Joint Gaussian Mixture Construction Based on a Diffeomorphism)**.** *Let $U \subseteq \mathbb{R}^d$ be a triangulated space with partition of unity $\{\phi_i\}_{i=1}^K$, and let p_U be a uniform PDF. The component weights are $w_i = \int_U \phi_i(u)p_U(u)\,\mathrm{d}u$. For each element T_i, a local Gaussian $\mathcal{N}(u|m_{U,i}, \Sigma_{U,i})$ is constructed, giving $\mathcal{G}(u|g_U) = \sum_i w_i\mathcal{N}(u|m_{U,i}, \Sigma_{U,i})$.*

For a diffeomorphism [1, 2] $f\colon U \to X$, let $J_i = J_f(m_{U,i})$. The local linearization of f induces $m_{X,i} = f(m_{U,i})$ and $\Sigma_{X,i} = J_i\Sigma_{U,i}J_i^\top$. The corresponding joint component is

$$\mathcal{N}\left(\begin{bmatrix}u\\x\end{bmatrix}\,\middle|\,\begin{bmatrix}m_{U,i}\\m_{X,i}\end{bmatrix}, \begin{bmatrix}\Sigma_{U,i} & \Sigma_{U,i}J_i^\top\\ J_i\Sigma_{U,i} & J_i\Sigma_{U,i}J_i^\top\end{bmatrix}\right),$$

and the joint GMM is obtained by mixing these components with the same weights,

$$\mathcal{G}(u, x|g_U, g_X) = \sum_{i=1}^K w_i\,\mathcal{N}\left(\begin{bmatrix}u\\x\end{bmatrix}\,\middle|\,\begin{bmatrix}m_{U,i}\\m_{X,i}\end{bmatrix}, \begin{bmatrix}\Sigma_{U,i} & \Sigma_{U,i}J_i^\top\\ J_i\Sigma_{U,i} & J_i\Sigma_{U,i}J_i^\top\end{bmatrix}\right).$$

Remark 0.1. *Thus, the joint representation is constructed in closed form from the uniform probability density function, discretization, and local Jacobians. Its marginals and conditionals are available analytically.*

Remark 0.2. *One can extend for arbitrary PDF on the parameter domain but for special choices like the uniform probability density functions over simplicial complexes [3] one can derive an analytical sharp bound for the global Kullback–Leibler divergence [4].*

Example 0.1 (Particle on a circular trajectory [5])**.** *Consider a particle on the trajectory $r(t) = (R\cos t, R\sin t)^\top$ with velocity $\dot{r}(t) = (-R\sin t, R\cos t)^\top$, $t \in [0, \pi/2]$. We construct local Gaussian components for the embeddings $u(t) = (t, r_1(t), r_2(t))$ and $x(t) =$*

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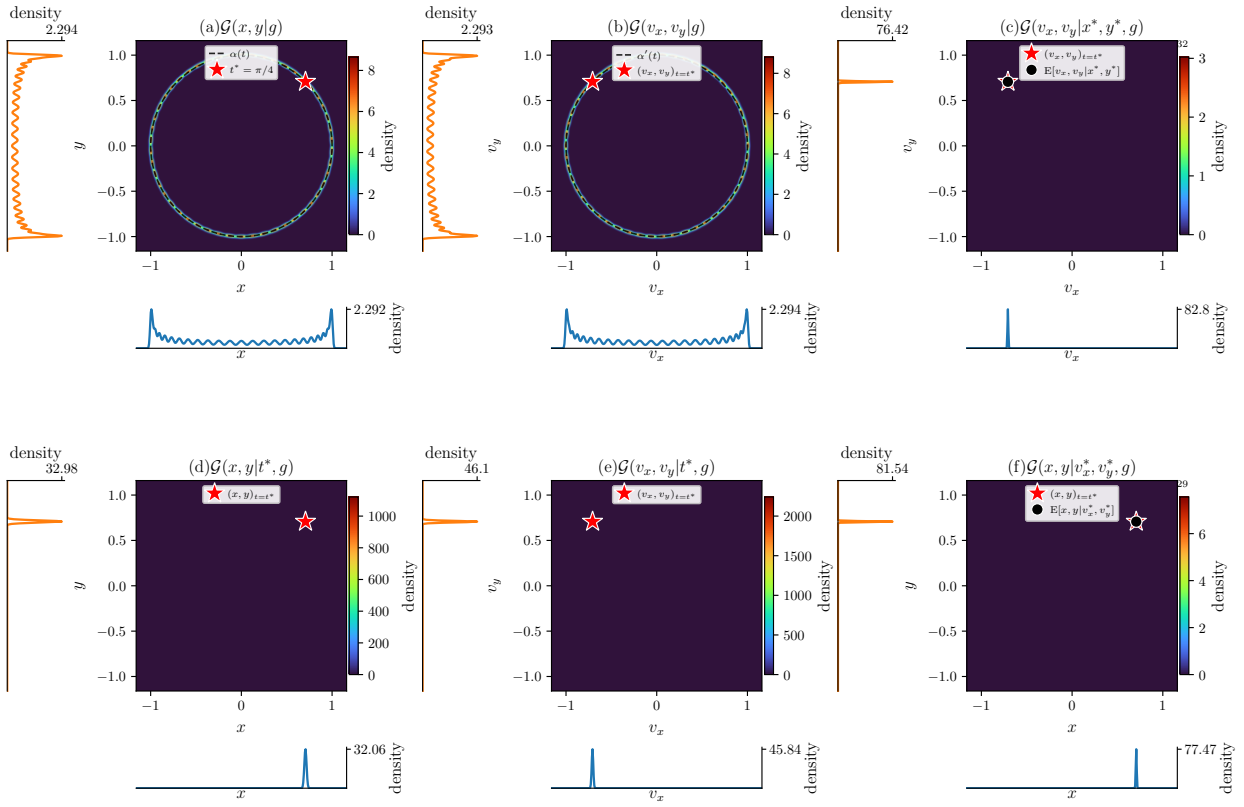


FIGURE 1. Joint GMM for a particle on a circular trajectory: (a,b) position and velocity marginals; (c) velocity conditioned on position; (d,e) position and velocity conditioned on t; (f) position conditioned on velocity.

$(1, \dot{r}_1(t), \dot{r}_2(t))$ *and couple them through the local Jacobians as described above. The resulting joint GMM provides the marginals* $q(r_1, r_2)$ *and* $q(\dot{r}_1, \dot{r}_2)$ *and the conditionals* $q(\dot{r}_1, \dot{r}_2|r_1^*, r_2^*)$ *and* $q(r_1, r_2|\dot{r}_1^*, \dot{r}_2^*)$*. Figures 1 and 2 illustrate these distributions together with the parameter-conditioned distributions. In particular, position can be inferred directly from velocity without explicitly recovering the underlying parameter* t*.*

Abstract

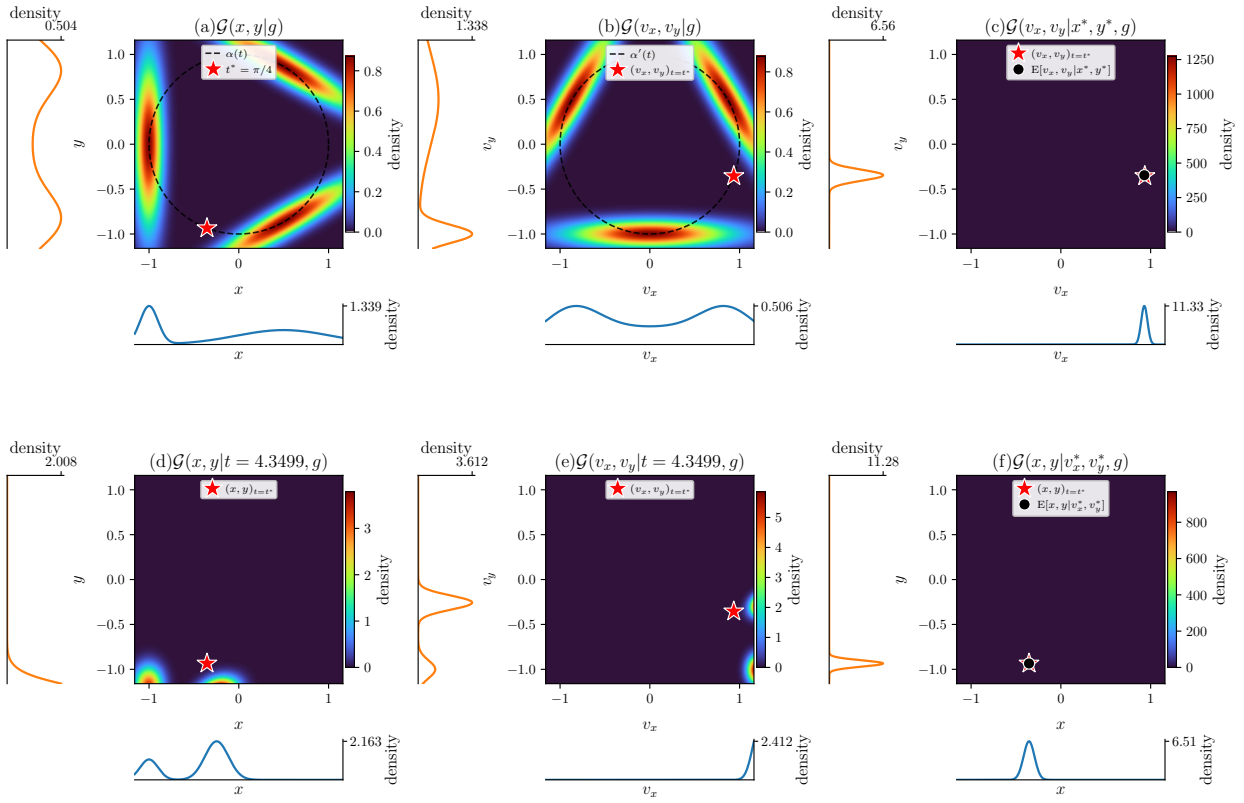


FIGURE 2. Joint GMM for a particle on a circular trajectory: (a,b) position and velocity marginals; (c) velocity conditioned on position; (d,e) position and velocity conditioned on t; (f) position conditioned on velocity.

Extract

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Ali Darijani[⊛□#¶⊥]
Research Staff, Fraunhofer IOSB of Fraunhofer-Gesellschaft, Germany
Research Staff, KIT Department of Computer Science, Germany
Alumnus, RWTH Aachen University, Germany
Alumnus, Sharif University of Technology, Iran
Email address: ali.darijani@iosb.fraunhofer.de
Email address: ali.darijani@kit.edu
Email address: ali.darijani@rwth-aachen.de
Email address: ali.darijani@alum.sharif.edu

Jürgen Beyerer[⊛⊥]
Professor, KIT Department of Computer Science, Germany
Director, Fraunhofer IOSB of Fraunhofer-Gesellschaft, Germany
Alumnus, Karlsruhe Institute of Technology, Germany
Email address: juergen.beyerer@kit.edu
Email address: juergen.beyerer@iosb.fraunhofer.de

Symbol Legend
⊛: Idea, □: Math, #: Programming, ¶: Writing, ⊥: Critique

Abstract